

people may be uncomfortable with robots replacing human beings, especially in very intimate settings such as hospitals.

Unemployment. Studies suggest a downward pressure on employment and wages if machines could entirely substitute workforce, especially low-skilled labour. This, in turn, may act as a deterrent to adoption of automation in a highly populated country like India.

Access to data. For companies that are looking to apply automation to any number of areas, data access is going to be one of the biggest challenges. Artificial intelligence and automation devices require hundreds of thousands times more information than humans to understand concepts or recognise features. Sometimes, small companies and startups may not have sufficient data collection facilities to run automation processes.

Moving forward

Today, manufacturing has evolved to such an extent that only through innovation and investment in technologies can businesses gain the productivity and efficiency advantage to compete in the global market.

A Frost & Sullivan whitepaper titled *Enhancing India's Manufacturing Competitiveness—Reality of Adopting Technologies and Trends* concedes that the next industrial revolution will be based on unification of the information amongst participants in the entire value chain—from product inception to design, manufacturing, services and even refurbishment. The report notes, "It will transform the manufacturing processes in sync with the speed of change in customer needs—which implies making the production process flexible without taking excess time."

Industrial automation will evolve in sync with these market pulses. In fact, industrial automation is ahead of most other industries in readiness for the Internet of Things (IoT) and, more specifically, the Industrial Internet of Things (IIoT) or Industry 4.0, which is one of the primary megatrends impacting the global automation market.

A recent report by HIS Technology

Emerging trends

1. Evolution of smart human-machine interface (HMI) to enable intelligent interaction and efficient data presentation
2. Use of vision systems in quality inspection to improve efficiency and precision level
3. Use of Big Data in manufacturing processes to improve productivity while reducing down time and aiding predictive maintenance
4. Industry 4.0, the Industrial Internet of Things (IIoT) enabling smart manufacturing with decentralised intelligence

forecasts industrial automation to make up majority of the IoT by 2025, with Internet protocol (IP)-addressable devices as the key enabler. The report states that industrial automation accounted for more than half of the installed base for all Internet-connected devices in 2012. The sector will grow at 36.3 per cent annually to account for nearly three-fourths of all connected devices by 2025.

The Industry 4.0 concept focuses on smart factories, connected machines, the IoT and industrial Internet. Interconnection of machinery, sensors and control systems together via intelligent networks helps manufacturers to achieve a slew of features. These include dynamic response to product demands, rapid new product manufacturing, real-time production and supply chain network optimisation, highly flexible production, strong product and mass customisation, and self-optimisation, self-configuration and self-diagnostics.

Today, the industrial automation concept is based on having one centralised 'brain' collecting information from manufacturing assets to facilitate production decisions. But in smart manufacturing, the intelligence is decentralised, with all of the smart assets within the factory having full information about themselves, and equipped with the processing power to optimise their own productivity and efficiency. These smart assets are also hyperconnected to optimise and coordinate each step of the manufacturing process. **EFY**

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